

AGILE SOFTWARE ENGINEERING (CS303)

User Stories & Acceptance Criteria

Learning Objectives

Understand Agile requirements, user stories, story structure, INVEST principle, acceptance criteria, and best practices for writing effective user stories.

Introduction to Agile Requirements

In Agile development, requirements are captured in a lightweight and flexible manner. Instead of lengthy requirement documents, Agile teams use User Stories to represent user needs.

What is a User Story?

A User Story is a short description of a feature from the perspective of an end user. It focuses on the value delivered to the customer rather than technical implementation.

Standard User Story Format

As a , I want so that .

Example User Story

As a student, I want to download course notes so that I can prepare for examinations offline.

Characteristics of Good User Stories

User stories should be simple, understandable, valuable, testable, and focused on user needs.

INVEST Principle

I - Independent
N - Negotiable
V - Valuable
E - Estimable
S - Small
T - Testable

Acceptance Criteria

Acceptance Criteria define the conditions that must be satisfied before a user story is considered complete. They provide clarity and help testers verify functionality.

Example Acceptance Criteria

User Story: Student Login

Acceptance Criteria:

1. Student can enter valid username and password.
2. System authenticates credentials.
3. Dashboard is displayed after successful login.
4. Error message is shown for invalid credentials.

Benefits of Acceptance Criteria

Provides clarity, improves communication, reduces misunderstandings, and ensures better testing coverage.

Good vs Bad User Stories

Good Story: Clear, concise, user-focused, testable.

Bad Story: Ambiguous, too large, technical, or lacking user value.

Common Mistakes

Writing overly large stories, missing acceptance criteria, focusing on implementation details, and ignoring business value.

Real World Applications

User stories are widely used in Scrum, Kanban, Jira, Azure DevOps, and other Agile project management tools.

Exam Important Questions

1. Define User Story.
2. Explain the standard format of a User Story.
3. Discuss the INVEST Principle.
4. What are Acceptance Criteria?
5. Differentiate good and bad User Stories.

Summary

User Stories and Acceptance Criteria are essential Agile requirement techniques. They help teams understand user needs, define success conditions, and deliver valuable software efficiently.